

LESSON check 8a-uniqueness

0001

We pick up at uniqueness in Problem 8(a) of the automorphisms worksheet, not Chapter 6. You skipped Chapter 6 for this sheet; Problems 7 to 10 are the plan, and 7(a) to 7(c) are already typeset.

Your existence argument for 8(a) is agreed. You built where and come from the evaluation maps. One half is left.

Problem 8(a). Let be irreducible, with roots and in (possibly different) extensions of . Show there is a unique isomorphism fixing and sending .

What you need:

- is your isomorphism: it fixes and sends .
- is any other isomorphism that fixes and sends .
- Every element of is for some (the evaluation map is surjective).
- A map that adds, multiplies and fixes commutes with evaluating a polynomial with coefficients in (the step inside your 7(b)).

Let fix and send . Show that .

YOUR ANSWER question 0001 · attempt 1 of 24 · tap to write

$(x \in F, \tau(x) = \sigma(x) \text{ is trivial})$

Let $x \in F(\alpha)$, $x = \sum_{i=0}^n a_i \alpha^i$, a_i

$$\tau(x) = \sum_{i=0}^n a_i \tau(x)^i \quad \text{through auto prop}$$

$$= \sum_{i=0}^n a_i \beta^i$$

$$= \sum_{i=0}^n a_i \sigma(x)^i = \sigma(x) \quad \text{agree}$$

you · 12:26 PM · revised

? LESSON check 8b-shift-p3

0002

Your uniqueness argument is right, and 8(a) is done. τ and σ both fix each a_i , both respect sums and powers, and both send $\alpha \mapsto \beta$, so they agree on every $a_i \alpha^i$. That is every element of $F(\alpha)$. The worksheet file now has all of 8(a) typeset: existence, then uniqueness.

Next is 8(b). It is the step 8(c) needs to get $[(\zeta_p):] = p-1$.

Problem 8(b). Let p be prime and $\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$.

Show that Φ_p is irreducible over $\mathbb{Q}$. Eisenstein does not apply

Eisenstein does not apply to Φ_p as it stands, because every coefficient is 1. The standard move is to substitute $x \mapsto x+1$ first. Try it on the smallest case before the general one.

What you need:

- $\Phi_3(x) = x^2 + x + 1$.
- Eisenstein at a prime q (yours from Chapter 5): if q divides every coefficient except the leading one, and q^2

does not divide the constant term, the polynomial is irreducible over .

Expand $\Phi_3(x+1)$. Does Eisenstein at $q=3$ apply to the result?

YOUR ANSWER question 0001 · attempt 2 of 24 · tap to write

$$\Phi_3(x) = x^2 + x + 1$$

$$\Phi_3(x+1) = (x+1)^2 + x+1 + 2$$

$$= x^2 + 2x + 1 + x + 1 + 2$$

$$= x^2 + 3x + 4$$

... Eisenstein doesn't apply here
if we had +4 instead of +

LESSON check 8b-general-shift

0003

Your instinct is right: the answer is $x^2 + 3x + 3$, and Eisenstein at 3 does apply. The +4 came from one slip. You shifted the constant term: $\Phi_3(x+1) = (x+1)^2 + (x+1) + 1$, and that last 1 stays 1, not 2.

$$\Phi_3(x+1) = x^2 + 2x + 1 + x + 1 + 1 = x^2 + 3x + 3.$$

3 divides 3 and 3, and 9 does not divide 3. So Φ_3 is irreducible.

Now the general prime. Expanding $\Phi_p(x+1)$ term by term is hopeless, so use a closed form instead.

Problem 8(b). Let p be prime and $\Phi_p(x) = x^{p-1} + \dots + x + 1$. Show that Φ_p is irreducible over $\mathbb{Q}$.

What you need:

- $(x-1)\Phi_p(x) = x^p - 1$ (multiply it out: everything telescopes).
- The binomial theorem: $(x+1)^p = \sum_{k=0}^p \binom{p}{k} x^k$.
- Eisenstein at q : q divides every coefficient but the leading one, and q^2 does not divide the constant term, so the polynomial is irreducible over $\mathbb{Q}$.
- If $f(x+1)$ is irreducible, so is $f(x)$ (a factorisation of one shifts to a factorisation of the other).

Put $x+1$ into $(x-1)\Phi_p(x) = x^p - 1$. Write $\Phi_p(x+1)$ as a polynomial whose coefficients are binomial coefficients $\binom{p}{k}$.

YOUR ANSWER question 0001 · attempt 3 of 24 · tap to write

$$\Phi_p(x) = x^{p-1} + \dots + x + 1$$

$$(x-1)\Phi_p(x) = x^p - 1$$

Plug in $x+1$: $(x+1)^p - 1$

$$= \left(\sum_{i=0}^p \binom{p}{i} x^i \right) - 1$$

$$= \sum_{i=1}^p \binom{p}{i} x^i$$

... This is reducible (common

LESSON check 8b-divide-by-x

0004

Your right-hand side is correct: $(x+1)^p - 1 = \sum_{i=1}^p \binom{p}{i} x^i$. And yes, it is reducible. It is supposed to be. That common x is the factor $x-1$ from the left-hand side, shifted.

You substituted into the right side only. Do the left side too:

$$((x+1) - 1) \Phi_p(x+1) = \sum_{i=1}^p \binom{p}{i} x^i.$$

The left side is $x \cdot \Phi_p(x+1)$. So the x you found on the right is exactly this x , and cancelling it leaves $\Phi_p(x+1)$ alone. That is the polynomial we test, not $(x+1)^p - 1$.

Problem 8(b). Let p be prime and $\Phi_p(x) = x^{p-1} + \dots + x + 1$. Show that Φ_p is irreducible over $\mathbb{Q}$.

What you need:

- $(x-1) \Phi_p(x) = x^p - 1$.
- Binomial theorem: $(x+1)^p = \sum_{k=0}^p \binom{p}{k} x^k$.
- Eisenstein at q : q divides every coefficient but the leading one, and q^2 does not divide the constant term, so the polynomial is irreducible over $\mathbb{Q}$.
- If $f(x+1)$ is irreducible, so is $f(x)$.

Divide both sides by x . Write $\Phi_p(x+1)$ as a sum of binomial coefficients times powers of x , with the index range correct, and name its constant term and its leading coefficient.

YOUR ANSWER question 0001 · attempt 4 of 24 · tap to write

$$\Phi_p(x) = x^{p-1} + \dots + x + 1$$

$$(x-1) \Phi_p(x) = x^p - 1$$

← Plug in $x+1$: $(x+1)^p - 1$

$$= \left(\sum_{i=0}^p \binom{p}{i} x^i \right) - 1$$

$$= \sum_{i=1}^p \binom{p}{i} x^i$$

$$= x \Phi_p(x)$$

$$\Rightarrow \Phi_p(x) = \sum_{i=1}^p \binom{p}{i} x^{i-1}$$

LESSON check 8b-p-divides-binomial

0005

Your expansion is right: leading coefficient 1, constant term $\binom{p}{1} = p$, and the index shift x^{i-1} is correct. One label is wrong. After dividing by x you wrote $\Phi_p(x)$ on the left. It is $\Phi_p(x+1)$, because you substituted $x+1$ on the left as well. The polynomial Eisenstein is applied to $\Phi_p(x+1)$.

"Irreducible by Eisenstein" still skips the actual check. The prime is p . The constant term is p , so p^2 does not divide it. What remains is the middle coefficients.

For $p=5$: $\binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4} = 5, 10, 10, 5$. All are divisible by 5. For $p=4$, which is not prime, $\binom{4}{2} = 6$ is not divisible by 4.

Problem 8(b). Let p be prime and $\Phi_p(x) = x^{p-1} + \dots + x + 1$. Show that Φ_p is irreducible over $\mathbb{Q}$.

What you need:

- Your line: $\Phi_p(x+1) = \sum_{i=1}^p \binom{p}{i} x^{i-1}$.
- $\binom{p}{k} = \frac{p!}{k!(p-k)!}$, an integer.

$$\kappa!(p-\kappa)!$$

- Eisenstein at q : q divides every coefficient except the leading one, and q^2 does not divide the constant term. Then the polynomial is irreducible over $\mathbb{Q}$.
- If $f(x+1)$ is irreducible, so is $f(x)$.

Show that p divides $\binom{p}{k}$ for every $0 < k < p$, and say where you used that p is prime.

YOUR ANSWER question 0001 · attempt 5 of 24 · tap to write

$$\Phi_p(x) = x^{p-1} + \dots + x + 1$$

$$(x-1)\Phi_p(x) = x^p - 1$$

← Plug in $x+1$: $(x+1)^p - 1$

$$= \left(\sum_{i=0}^p \binom{p}{i} x^i \right) - 1$$

$$= \sum_{i=1}^p \binom{p}{i} x^i$$

$$= x \Phi_p(x+1)$$

$$\Rightarrow \Phi_p(x+1) = \sum_{i=1}^p \binom{p}{i} x^{i-1}$$

$$= x^{p-1} + \binom{p}{p-1} x^{p-2} + \dots + 1$$

LESSON check 8c-degree-p5

0006

Problem 8(b) is agreed. The label is fixed, you checked all three Eisenstein conditions at p , and you shifted back from $\Phi_p(x+1)$ to $\Phi_p(x)$. The line $p \mid \binom{p}{k}$ for $2 \leq k \leq p-1$ stands as a stated fact. The worksheet now has 8(b) typeset beside your handwriting. 8(a) is already done uniqueness included

handwriting. 8(a) is already done, uniqueness included.

Problem 8(c). Deduce that $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$ and that the injection of Problem 7(c) is an isomorphism, $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$.

What you need:

- $\zeta_p = e^{2\pi i/p}$, a primitive p -th root of unity.
- $\Phi_p(x) = x^{p-1} + \cdots + x + 1$, irreducible over $\mathbb{Q}$ by your 8(b).
- $(x - 1)\Phi_p(x) = x^p - 1$, your first line in 8(b).
- $[F(\alpha) : F] = \deg$ of the minimal polynomial of α over F .
- 7(c): $\sigma \mapsto k \bmod p$, where $\sigma(\zeta_p) = \zeta_p^k$, is an injective homomorphism $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/p\mathbb{Z})^\times$.
- 8(a): if α, β are roots of the same irreducible $q \in F[x]$, there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.

First, one small case. Take $p = 5$. What is the minimal polynomial of ζ_5 over $\mathbb{Q}$, and so what is $[\mathbb{Q}(\zeta_5) : \mathbb{Q}]$? Say why ζ_5 is a root of it.

YOUR ANSWER question 0001 · attempt 6 of 24 · tap to write

we DID this. Are
you high? (out back

kind: lesson title: 8(a) is done -- the open question is 8(c)

You are right: 8(a) is finished, uniqueness included, and I should not have put it in front of you again. It is typeset in the worksheet beside your handwriting (the 8a page). The last card was mislabelled "8a-uniqueness-again", but its question was 8(c). Nothing about 8(a) is being asked.

Problem 8(c). Deduce that $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$ and that the injection of Problem 7(c) is an isomorphism, $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$.

What you need:

- $\zeta_p = e^{2\pi i/p}$, a primitive p -th root of unity.
- $\Phi_p(x) = x^{p-1} + \cdots + x + 1$, irreducible over $\mathbb{Q}$ by your 8(b).
- $(x - 1)\Phi_p(x) = x^p - 1$, your first line in 8(b).
- $[F(\alpha) : F] = \deg$ of the minimal polynomial of α over F .
- 7(c): $\sigma \mapsto k \bmod p$, where $\sigma(\zeta_p) = \zeta_p^k$, is an injective homomorphism $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/p\mathbb{Z})^\times$.
- 8(a): if α, β are roots of the same irreducible $q \in F[x]$, there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.

First, the small case. Take $p = 5$. What is the minimal polynomial of ζ_5 over $\mathbb{Q}$, and so what is $[\mathbb{Q}(\zeta_5) : \mathbb{Q}]$? Say why ζ_5 is a root of it.

$$\zeta_p \text{ over } \mathbb{Q}$$

$$\downarrow$$

$$e^{2\pi i/p}$$

$$x^p - 1 = \underbrace{\Phi_p(x)}_{\text{so clearly}} (x-1) \dots$$

so clearly

$\Phi_p(x)$ must
have ζ_p as a
root. And Φ_p is
irreducible, so $[\mathbb{Q}(\zeta_p) : \mathbb{Q}]$

LESSON check 8c-surjective-roots

0008

kind: lesson title: 8(c) degree agreed -- now surjectivity

Your degree argument is right: $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$. The "clearly" hides one fact, which I take as read: $\zeta_p \neq 1$, so it is a root of the factor Φ_p rather than of $x - 1$. Monic, irreducible, and killing ζ_p makes Φ_p the minimal polynomial.

Problem 8(c), second half. Show the injection of 7(c) is onto, so $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$.

What you need:

- $\zeta_p = e^{2\pi i/p}$, with $\zeta_p^p = 1$ and $\zeta_p^k \neq 1$ for $1 \leq k \leq p - 1$.
- $\Phi_p(x) = x^{p-1} + \dots + x + 1$, irreducible over $\mathbb{Q}$ (8(b)), and $(x - 1)\Phi_p(x) = x^p - 1$.

- 7(c): $\sigma \mapsto k \bmod p$, where $\sigma(\zeta_p) = \zeta_p^k$, is an injective homomorphism into $(\mathbb{Z}/p\mathbb{Z})^\times$.
- 8(a): if α, β are roots of the same irreducible $q \in F[x]$, there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.

One piece first. Fix k with $1 \leq k \leq p-1$. Why is ζ_p^k a root of Φ_p , and what is $\mathbb{Q}(\zeta_p^k)$ as a field?

YOUR ANSWER question 0001 · attempt 8 of 24 · tap to write

Because ζ_p^k is not a
root of $x-1$, since $k >$
 $\mathbb{Q}(\zeta_p^k)$ is polynomials
 $\mathbb{Q}$ evaluated at ζ_p^k

LESSON check 8c-generator-power

0009

The root half is right, with one correction. The reason $\zeta_p^k \neq 1$ is $k < p$, not $k > 0$: at $k = p$ you get $\zeta_p^p = 1$. So $(\zeta_p^k)^p = (\zeta_p^p)^k = 1$, and since $\zeta_p^k \neq 1$ it is a root of the factor Φ_p , not of $x-1$.

The field half describes the elements of $\mathbb{Q}(\zeta_p^k)$ but does not say which field it is. The point is that it equals $\mathbb{Q}(\zeta_p)$. One

inclusion is free: ζ_p^k is already in $\mathbb{Q}(\zeta_p)$. The other needs ζ_p to be a polynomial in ζ_p^k — in fact a single power of it.

Try it on the smallest case first. Take $p=5$ and $k=2$.

What you need:

- $\zeta_5 = e^{2\pi i/5}$, so $\zeta_5^5 = 1$ and exponents of ζ_5 only matter mod 5.
- $\mathbb{Q}(\beta)$ is the smallest field containing $\mathbb{Q}$ and β ; it contains every power of β .

With $p=5$, $k=2$: find m with $(\zeta_5^2)^m = \zeta_5$. What condition on m , stated mod 5, did you use?

YOUR ANSWER question 0001 · attempt 9 of 24 · tap to write

This is worthless.

My proof of

$$[\mathbb{Q}(\zeta_p) : \mathbb{Q}] =$$

is good enough

us on

Moving on. 8(c) goes into the write-up with your degree argument. The onto half is marked there as not yet proved, not filled in by me. That is the one gap left in Problem 8.

Problem 9(a). Let $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$. Show that ζ is a root of $x^4 + 1$, that $x^4 + 1$ is irreducible over $\mathbb{Q}$, and hence that $[K : \mathbb{Q}] = 4$.

This is 8(b) and 8(c) again on a new polynomial, so it is the whole part, not a warm-up.

What you need:

- $\zeta_8 = e^{2\pi i/8} = e^{\pi i/4}$.
- Eisenstein at a prime q : monic, q divides every non-leading coefficient, q^2 does not divide the constant term. Then it is irreducible over $\mathbb{Q}$.
- $f(x)$ is irreducible exactly when $f(x+c)$ is, for any $c \in \mathbb{Q}$ (your 8(b) shift).
- $[F(\alpha) : F]$ is the degree of the minimal polynomial of α over F .

Show ζ_8 is a root of $x^4 + 1$, that $x^4 + 1$ is irreducible over $\mathbb{Q}$, and so that $[\mathbb{Q}(\zeta_8) : \mathbb{Q}] = 4$.

YOUR ANSWER question 0001 · attempt 10 of 24 · tap to write

NO! we can finish
onto gap, but the
you were marking
were just you being
about my $[\mathbb{Q}(\zeta_8) : \mathbb{Q}]$

That part is done.
to the rest of 8
no idea what I'm

LESSON check 8c-onto-statement

0011

Back to 8(c), then. The degree half stays as you wrote it. What is left is one claim: every k from 1 to $p-1$ is actually hit.

What "onto" means here. Problem 7(c) gave you a map $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$. It sends σ to the k with $\sigma(\zeta_p) = \zeta_p^k$. You proved it is injective. Onto says the reverse: pick any k with $1 \leq k \leq p-1$, and some automorphism sends ζ_p to ζ_p^k . So you must build, for each such k , an automorphism σ_k of $\mathbb{Q}(\zeta_p)$ fixing $\mathbb{Q}$ with $\sigma_k(\zeta_p) = \zeta_p^k$.

$$\psi: \text{Aut} \rightarrow \mathbb{Z}_{\text{mod } p}$$

The tool is your own 8(a). It needs two things about ζ_p^k : that it has the same minimal polynomial as ζ_p , and that $\mathbb{Q}(\zeta_p^k)$ is the same field as $\mathbb{Q}(\zeta_p)$.

What you need:

- $\zeta_p = e^{2\pi i/p}$, p prime.
- $\Phi_p(x) = x^{p-1} + \dots + x + 1$, the minimal polynomial of ζ_p over $\mathbb{Q}$ (your degree half).
- 8(a): if α, β have the same minimal polynomial over F , there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.
- 7(c): the injection $\sigma \mapsto k$, where $\sigma(\zeta_p) = \zeta_p^k$.

Prove: for each k with $1 \leq k \leq p-1$ there is an automorphism σ_k of $\mathbb{Q}(\zeta_p)$ fixing $\mathbb{Q}$ with $\sigma_k(\zeta_p) = \zeta_p^k$. Hence the injection of 7(c) is onto, so $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$.

Let $k \in \mathbb{Z}_{\text{mod } p}$

Show $\exists \sigma \in \text{Aut}(\mathbb{Q}(\zeta_p)) : \mathbb{Q}$

$$\sigma(\zeta_p) = \zeta_p^k \dots$$

I'm stuck ...

LESSON check 8c-onto-first-piece

0012

Your statement is exactly right. Now use it: 8(a) turns it into two smaller facts about ζ_p^k , and you prove the first one now.

8(a) builds an isomorphism $\mathbb{Q}(\zeta_p) \rightarrow \mathbb{Q}(\zeta_p^k)$ sending $\zeta_p \mapsto \zeta_p^k$. It needs ζ_p and ζ_p^k to have the same minimal polynomial. That is the first fact. The second fact, $\mathbb{Q}(\zeta_p^k) = \mathbb{Q}(\zeta_p)$, is what makes the isomorphism an automorphism. It comes next turn.

For the first fact you only need $\Phi_p(\zeta_p^k) = 0$. Then Φ_p is irreducible, monic, and kills ζ_p^k , so it is the minimal polynomial of ζ_p^k too.

What you need:

- $\zeta = e^{2\pi i/p}$ n prime $1 < k < n-1$

- $\Phi_p(x) = x^{p-1} + \dots + x + 1$, irreducible over $\mathbb{Q}$ (your degree half).
- $(x-1)\Phi_p(x) = x^p - 1$.
- 8(a): if α, β have the same minimal polynomial over F , there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.

Show: $\Phi_p(\zeta_p^k) = 0$ for each k with $1 \leq k \leq p-1$.

YOUR ANSWER question 0001 · attempt 12 of 24 · tap to write

Let $k \in \mathbb{Z}_{\text{mod } p}$

Show $\exists \sigma \in \text{Aut}(\mathbb{Q}(\zeta_p) : \mathbb{Q})$

$$\sigma(\zeta_p) = \zeta_p^k \dots$$

Note $\Phi_p(\zeta_p^k) = x^{p-1} + \dots + 1 \Big|_{x = e^{2\pi i/p^k}} \quad (0 < k < p)$

I don't know how to

LESSON check 8c-onto-plug-into-identity

0013

Do not expand the sum. Plug ζ_p^k into the product identity instead: it has Φ_p as a factor, so it hands you $\Phi_p(\zeta_p^k)$ times something.

Your setup is right: $\zeta_p^k = e^{2\pi i k/p}$ with $0 < k < p$. Summing p powers

of it directly is the hard road. The identity $(x-1)\Phi_p(x) = x^p - 1$ is the easy one.

Put $x = \zeta_p^k$ into it. Two things to settle:

- What is $(\zeta_p^k)^p$? So what is the right-hand side?
- Is $\zeta_p^k - 1$ zero or not, given $0 < k < p$?

Then a product of two complex numbers is 0 with one factor nonzero. That finishes it.

What you need:

- $\zeta_p = e^{2\pi i/p}$, p prime, $1 \leq k \leq p-1$.
- $\Phi_p(x) = x^{p-1} + \dots + x + 1$.
- $(x-1)\Phi_p(x) = x^p - 1$, true for every complex x .
- $e^{2\pi i t} = 1$ exactly when t is an integer.

Show: $\Phi_p(\zeta_p^k) = 0$ for each k with $1 \leq k \leq p-1$, by evaluating $(x-1)\Phi_p(x) = x^p - 1$ at $x = \zeta_p^k$.

YOUR ANSWER question 0001 · attempt 13 of 24 · tap to write

$$\text{Let } k \in \mathbb{Z}_{\text{mod } p}$$

$$\text{Show } \exists \sigma \in \text{Aut}(\mathbb{Q}(\zeta_p) : \mathbb{Q})$$

$$\sigma(\zeta_p) = \zeta_p^k \dots$$

$$\text{Consider } x^p - 1 = (x-1)\Phi_p(x)$$

$$\underbrace{\quad}_{\sim} (\zeta_p^k)^p - 1 = \zeta_p^k - 1 = 1 - 1 = 0$$

But $\sum_p^k \neq 1$ since 0

LESSON check 8c-onto-same-field

0014

Step one is done: $\Phi_p(\zeta_p^k) = 0$. One line of it is wrong, though.

You wrote $(\zeta_p^k)^p - 1 = \zeta_p^k - 1 = 1 - 1$. That says $\zeta_p^k = 1$, which your next line correctly denies. The right middle step is $(\zeta_p^k)^p = (\zeta_p^p)^k = 1^k = 1$. So the right-hand side is 0, $\zeta_p^k - 1 \neq 0$, and $\Phi_p(\zeta_p^k) = 0$. Your conclusion stands.

So ζ_p^k is a root of the irreducible Φ_p , just as ζ_p is. Problem 8(a) then gives an isomorphism $\mathbb{Q}(\zeta_p) \rightarrow \mathbb{Q}(\zeta_p^k)$ sending $\zeta_p \mapsto \zeta_p^k$. It is an automorphism only if those two fields are the same field. That is step two.

Clearly $\mathbb{Q}(\zeta_p^k) \subseteq \mathbb{Q}(\zeta_p)$. The other inclusion needs $\zeta_p \in \mathbb{Q}(\zeta_p^k)$.

Because
autom

What you need:

- $\zeta_p = e^{2\pi i/p}$, p prime, $1 \leq k \leq p-1$.
- $\zeta_p^p = 1$, so exponents of ζ_p only matter mod p .
- $p \nmid k$, so k is a unit in $\mathbb{Z}/p$: some integer m has $km \equiv 1 \pmod{p}$.
- $\mathbb{Q}(\beta)$ is the smallest field containing $\mathbb{Q}$ and β , so it contains every power of β .

Show: $\zeta_p \in \mathbb{Q}(\zeta_p^k)$, by writing ζ_p as a power of ζ_p^k . Conclude $\mathbb{Q}(\zeta_p^k) = \mathbb{Q}(\zeta_p)$.

YOUR ANSWER question 0001 · attempt 14 of 24 · tap to write

Let $k \in \mathbb{Z}_{\text{mod } p}$

Show $\exists \sigma \in \text{Aut}(\mathbb{Q}(\zeta_p) : \mathbb{Q})$ st.

$\sigma(\zeta_p) = \zeta_p^k \dots$

Consider $x^p - 1 = (x-1)\Phi_p(x)$

$$\underbrace{\quad}_{\sim} (\zeta_p^k)^p - 1 = \zeta_p^k - 1 = 1 - 1 = 0$$

But $\zeta_p^k \neq 1$ since $0 < k < p$,

so $\zeta_p^k - 1 \neq 0$. Hence $\Phi_p(\zeta_p^k)$

must be zero $\therefore$

So $\sigma(\zeta_p) = \zeta_p^k$ is an automorphism if

$\gcd(k, p) = 1$ Clearly $\mathbb{Q}(\zeta_p^k) \subseteq \mathbb{Q}(\zeta_p)$.

LESSON check 9a-x4-plus-1-pose

0015

Step two is right, and with it 8(c) is finished: the injection $\text{Aut}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \rightarrow (\mathbb{Z}/p)^\times$ is onto. No question on the page.

Your k' with $kk' \equiv 1 \pmod{p}$ gives $(\zeta_p^k)^{k'} = \zeta_p^{kk'} = \zeta_p$, so $\zeta_p \in \mathbb{Q}(\zeta_p^k)$ and the fields are equal. Then 8(a) makes $\zeta_p \mapsto \zeta_p^k$ an automorphism, for every k in $(\mathbb{Z}/p)^\times$.

Your middle line still reads $\zeta_p^k - 1 = 1 - 1$. The write-up says $(\zeta_p^k)^p = (\zeta_p^p)^k = 1$ instead, and 8(c) is now typeset in full.

Problem 9(a). Let $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$. Show that ζ is a root of $x^4 + 1$, that $x^4 + 1$ is irreducible over $\mathbb{Q}$, and hence that $[K:\mathbb{Q}] = 4$.

What you need:

- $\zeta_8 = e^{2\pi i/8} = e^{\pi i/4}$.
- Eisenstein at a prime q : monic, q divides every non-leading coefficient, q^2 does not divide the constant term. Then it is irreducible over $\mathbb{Q}$.
- $f(x)$ is irreducible exactly when $f(x+c)$ is, for any $c \in \mathbb{Q}$ (your 8(b) shift).
- $[F(\alpha):F]$ is the degree of the minimal polynomial of α over F .

Show ζ_8 is a root of x^4+1 , that x^4+1 is irreducible over $\mathbb{Q}$, and so that $[\mathbb{Q}(\zeta_8):\mathbb{Q}] = 4$.

YOUR ANSWER question 0001 · attempt 15 of 24 · tap to write

$$\zeta_8 = e^{2\pi i/8}$$

$$x^4+1 \Big|_{x=\zeta_8} = e^{8\pi i/8} + 1 = e^{i\pi} + 1 = -1 + 1 = 0$$

Further x^4+1 has only ± 1 as possible rational roots, and neither work. So it is clearly irreducible over $\mathbb{Q}$. So x^4+1 is minimal polynomial of ζ_8 , so $[\mathbb{Q}(\zeta_8):\mathbb{Q}] = 4 \therefore$

LESSON check 9a-irreducible-quadratics

0016

The root is right: $\zeta_8^4 = e^{i\pi} = -1$, so $\zeta_8^4 + 1 = 0$. The irreducibility is not proved. No question on the page.

"No rational roots" only rules out a linear factor. A quartic can still split into two quadratics. It happens: x^4+4 has no rational roots, yet

$$x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2).$$

So x^4+1 needs an argument that kills a quadratic-times-quadratic factorisation too. Rational-root testing settles

irreducibility only in degree 2 and 3.

Problem 9(a), irreducibility. Show that x^4+1 is irreducible over $\mathbb{Q}$.

What you need:

- Eisenstein at a prime q : monic, q divides every non-leading coefficient, q^2 does not divide the constant term. Then it is irreducible over $\mathbb{Q}$.
- $f(x)$ is irreducible exactly when $f(x+c)$ is, for any $c \in \mathbb{Q}$ (your 8(b) shift).
- Gauss: a monic integer polynomial that factors over $\mathbb{Q}$ factors into monic integer polynomials.

Your root computation and the step "minimal polynomial has degree 4, so $[\mathbb{Q}(\zeta_8) : \mathbb{Q}] = 4$ " stand once this is done.

Prove that x^4+1 is irreducible over $\mathbb{Q}$ -- ruling out two quadratic factors, not only linear ones.

YOUR ANSWER question 0001 · attempt 16 of 24 · tap to write

$$\zeta_8 = e^{2\pi i/8}$$

$$x^4+1 \Big|_{x=\zeta_8} = e^{8\pi i/8} + 1 = e^{i\pi} + 1 = -1 + 1 = 0$$

$$(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$$

$$(x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2, \text{ irreducible by Eisenstein's k=2}$$

So x^4+1 is irreducible. So x^4+1 is minimal polynomial of

$$\zeta_8, \text{ so } [\mathbb{Q}(\zeta_8) : \mathbb{Q}] = 4 \therefore$$

LESSON check 9b-quadratic-factors

0017

9(a) is right and complete. $(x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$ is Eisenstein at 2: 2 divides 4, 6, 4, 2 and $4 \nmid 2$. So $x^4 + 1$ is irreducible, it is the minimal polynomial of ζ_8 , and $[\mathbb{Q}(\zeta_8) : \mathbb{Q}] = 4$. No question on the page; the bottom-right " $p \in \mathbb{Q}(\zeta_p^k)$ " is left over from 8(c). The write-up now has 9(a) as you wrote it.

So $x^4 + 1$ has no factor over $\mathbb{Q}$. Over a slightly bigger field it does.

Problem 9(b). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, show that $x^4 + 1$ factors into quadratics over each of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$.

What you need:

- $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, and likewise $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-2})$.
- "Factors into quadratics over F ": $x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d)$ with $a, b, c, d \in F$.
- An explicit factorisation, multiplied out, is a complete proof. No irreducibility argument is asked for.

Write $x^4 + 1$ as a product of two quadratics with coefficients in $\mathbb{Q}(\sqrt{2})$, then in $\mathbb{Q}(i)$, then in $\mathbb{Q}(\sqrt{-2})$.

YOUR ANSWER question 0001 · attempt 17 of 24 · tap to write

$$\zeta_8 = e^{2\pi i/8}$$

$$x^4 + 1, \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(i), \quad \mathbb{Q}(\sqrt{-2})$$

$$\begin{array}{ccc} & \downarrow & \downarrow \\ \underbrace{(x^2 + \sqrt{2})(x^2 - \sqrt{2}) + 1} & & (x^2 + i)(x^2 - i) \end{array}$$

In
math

STUCK...
what's the
clever trick?

LESSON check 9b-difference-of-squares

0018

The trick is to add and subtract a middle term so $x^4 + 1$ becomes a difference of two squares. Then $A^2 - B^2 = (A - B)(A + B)$ does the rest.

You already did this over $\mathbb{Q}(i)$, and it is right: $x^4 + 1 = x^4 - i^2 = (x^2 + i)(x^2 - i)$. That case is done.

Your $\mathbb{Q}(\sqrt{2})$ attempt, $(x^2 + \sqrt{2})(x^2 - \sqrt{2}) + 3$, is true but is not a product. The $+3$ is the problem. Using $\sqrt{2}$ on the constant term only gives $x^4 - 2$, the wrong constant. Put $\sqrt{2}$ on the x term instead:

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2.$$

Here $2x^2$ is a square once $\sqrt{2}$ is allowed. For $\mathbb{Q}(\sqrt{-2})$, subtract instead of add:

$$x^4 + 1 = (x^2 - 1)^2 + 2x^2.$$

Ask which number squares to -2 .

Problem 9(b). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, show that $x^4 + 1$ factors into quadratics over each of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$.

What you need:

- $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$, and likewise $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-2})$.

- "Factors into quadratics over F ": $x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d)$ with $a, b, c, d \in F$.
- An explicit factorisation, multiplied out, is a complete proof.
- Difference of squares: $A^2 - B^2 = (A - B)(A + B)$.
- Done: over $\mathbb{Q}(i)$, $x^4 + 1 = (x^2 + i)(x^2 - i)$.

Use the two identities above to write $x^4 + 1$ as a product of two quadratics with coefficients in $\mathbb{Q}(\sqrt{2})$, and then in $\mathbb{Q}(\sqrt{-2})$.

YOUR ANSWER question 0001 · attempt 18 of 24 · tap to write

$$\zeta_8 = e^{2\pi i/8}$$

$$x^4 + 1, \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(i), \quad \mathbb{Q}(\sqrt{-2})$$

$$(x^2 + i)(x^2 - i)$$

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2$$

$$= (x^2 + 1)^2 - (\sqrt{2}x)^2$$

$$= (x^2 + 1 - \sqrt{2}x)(x^2 + 1 + \sqrt{2}x)$$

$$x^4 + 1 = (x^2 - 1)^2 + 2x^2$$

$$= (x^2 - 1)^2 - (\sqrt{-2}x)^2$$

$$= (x^2 - 1 + \sqrt{-2}x)(x^2 - 1 - \sqrt{-2}x)$$

LESSON check 9b-sqrt-minus-2-sign

0019

Your $\mathbb{Q}(\sqrt{2})$ factorisation is right, and so is $\mathbb{Q}(i)$. The $\mathbb{Q}(\sqrt{-2})$ one has a sign slip: your boxed product multiplies out to $x^4 + 4x^2 + 1$. not $x^4 + 1$.

Your first line is right: $x^4 + 1 = (x^2 - 1)^2 + 2x^2$. On the second line, $x^2 - 1$ turned into $x^2 + 1$. Nothing justifies that change. The $\sqrt{-2}$ belongs only on the x term, since $(\sqrt{-2}x)^2 = -2x^2$, so $+2x^2 = -(\sqrt{-2}x)^2$. The bracket $(x^2 - 1)$ stays exactly as it was.

Problem 9(b). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, show that $x^4 + 1$ factors into quadratics over each of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-2})$.

What you need:

- $\mathbb{Q}(\sqrt{-2}) = \{a + b\sqrt{-2} : a, b \in \mathbb{Q}\}$, and $(\sqrt{-2})^2 = -2$.
- "Factors into quadratics over F ": $x^4 + 1 = (x^2 + ax + b)(x^2 + cx + d)$ with $a, b, c, d \in F$.
- Difference of squares: $A^2 - B^2 = (A - B)(A + B)$.
- Done: $x^4 + 1 = (x^2 + i)(x^2 - i)$ over $\mathbb{Q}(i)$.
- Done: $x^4 + 1 = (x^2 + 1 - \sqrt{2}x)(x^2 + 1 + \sqrt{2}x)$ over $\mathbb{Q}(\sqrt{2})$.

Starting from $x^4 + 1 = (x^2 - 1)^2 - (\sqrt{-2}x)^2$, write $x^4 + 1$ as a product of two quadratics with coefficients in $\mathbb{Q}(\sqrt{-2})$.

YOUR ANSWER question 0001 · attempt 19 of 24 · tap to write

$$\zeta_8 = e^{2\pi i/8}$$

$$x^4 + 1, \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(i), \quad \mathbb{Q}(\sqrt{-2})$$

$$(x^2 + i)(x^2 - i)$$

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2$$

$$= (x^2 + 1)^2 - (\sqrt{2}x)^2$$

$$= (x^2 + 1 - \sqrt{2}x)(x^2 + 1 + \sqrt{2}x)$$

$$x^4 + 1 = (x^2 - 1)^2 + 2x^2$$

$$= (x^2 - 1)^2 - (\sqrt{-2}x)^2$$

$$= (x^2 - 1 + \sqrt{-2}x)(x^2 - 1 - \sqrt{-2}x)$$

LESSON check 9c-sqrt2-in-K

0020

All three factorisations in 9(b) are right now, including $\mathbb{Q}(\sqrt{-2})$: $(x^2 - 1 + \sqrt{-2}x)(x^2 - 1 - \sqrt{-2}x) = (x^2 - 1)^2 + 2x^2 = x^4 + 1$. 9(b) is in the write-up with your three lines, and the PDF builds.

Next is 9(c). It has two halves: $\mathbb{Q}(i, \sqrt{2}) \subseteq K$ needs i and $\sqrt{2}$ written in terms of ζ , and $K \subseteq \mathbb{Q}(i, \sqrt{2})$ needs ζ written in terms of i and $\sqrt{2}$. The hard half is getting $\sqrt{2}$ out of ζ , so do that one first.

Problem 9(c). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, show that $K = \mathbb{Q}(i, \sqrt{2})$.

What you need:

- $\zeta = \zeta_8 = e^{2\pi i/8}$, so $\zeta^8 = 1$ and $\zeta^4 = -1$.
- $K = \mathbb{Q}(\zeta)$ is every polynomial in ζ with rational coefficients.
- $\mathbb{Q}(i, \sqrt{2})$ is the smallest field containing $\mathbb{Q}$, i and $\sqrt{2}$.
- Two fields are equal when each contains the other's generators.
- Euler: $e^{i\theta} = \cos \theta + i \sin \theta$, and $\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$.

Write $\sqrt{2}$ as a polynomial in ζ with rational coefficients.

YOUR ANSWER question 0001 · attempt 20 of 24 · tap to write

$$K = \mathbb{Q}(\zeta_8), \text{ show } K = \mathbb{Q}(i, \sqrt{2})$$

$$\zeta_8 = e^{2\pi i/8} = e^{i\pi/4} = \cos(\pi/4) + i\sin(\pi/4)$$

$$\begin{aligned} \sqrt{2} &= \zeta_8 + (\zeta_8)^{-1} = \cos(\pi/4) + \cos(-\pi/4) + i\sin(\pi/4) + i\sin(-\pi/4) \\ &= \sqrt{2}/2 + \sqrt{2}/2 + i\sqrt{2}/2 - i\sqrt{2}/2 \\ &= \sqrt{2} \end{aligned}$$

certainly present in $\mathbb{Q}(\zeta_8)$ (✓)

LESSON check 9c-repose

0021

Right: $\sqrt{2} = \zeta + \zeta^{-1}$, and the cosines add while the sines cancel. It sits in K because $\zeta^{-1} = \zeta^7$, a polynomial in ζ . With $i = \zeta^2$, the half $\mathbb{Q}(i, \sqrt{2}) \subseteq K$ is done.

What is left is the other inclusion: ζ written in terms of i and $\sqrt{2}$. That is the whole of 9(c) now, so here it is in full.

Problem 9(c). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, show that $K = \mathbb{Q}(i, \sqrt{2})$.

What you need:

- $\zeta = \zeta_8 = e^{2\pi i/8} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$, so $\zeta^8 = 1$ and $\zeta^4 = -1$.
- $K = \mathbb{Q}(\zeta)$ is every polynomial in ζ with rational coefficients.
- $\mathbb{Q}(i, \sqrt{2})$ is the smallest field containing $\mathbb{Q}$, i and $\sqrt{2}$.
- Two fields are equal when each contains the other's generators.
- $\cos \frac{\pi}{4} = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$.
- Already shown: $\sqrt{2} = \zeta + \zeta^{-1} = \zeta + \zeta^7$ and $i = \zeta^2$, so $\mathbb{Q}(i, \sqrt{2}) \subseteq K$.

Show that $K = \mathbb{Q}(i, \sqrt{2})$: write ζ in terms of i and $\sqrt{2}$, and conclude both inclusions.

$K = \mathbb{Q}(\zeta_8)$, show $K = \mathbb{Q}(i, \sqrt{2})$

$$\zeta_8 = e^{2\pi i/8} = e^{i\pi/4} = \cos(\pi/4) + i\sin(\pi/4) = \sqrt{2}/2 + i\sqrt{2}/2 \in \mathbb{Q}(i, \sqrt{2})$$

(coefficients both $1/2$) $\therefore$

LESSON check 9d-existence

0022

Right, and 9(c) is done. $\zeta = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$ puts ζ in $\mathbb{Q}(i, \sqrt{2})$, so $K \subseteq \mathbb{Q}(i, \sqrt{2})$. With last card's $\sqrt{2} = \zeta + \zeta^7$ and $i = \zeta^2$, the two fields are equal. The write-up now says exactly that.

Problem 9(d). With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$: for $k \in \{3, 5, 7\}$, show there is an automorphism σ_k of K with $\sigma_k(\zeta) = \zeta^k$, that each σ_k has order 2, and that the subfields of K fixed by σ_5 , σ_7 and σ_3 are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$ respectively.

Three pieces: existence, order 2, fixed fields. Existence first. It is the same move you made for σ_k in 8(c).

What you need:

- $\zeta = e^{2\pi i/8}$, so $\zeta^8 = 1$ and $\zeta^4 = -1$.
- $x^4 + 1$ is the minimal polynomial of ζ over $\mathbb{Q}$ (9(a)), and $[K : \mathbb{Q}] = 4$.
- 8(a): if α, β have the same minimal polynomial over F

• 9(a). If α, β have the same minimal polynomial over F , there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F and sending $\alpha \mapsto \beta$.

- An automorphism of K is an isomorphism $K \rightarrow K$.
- Order 2: $\sigma_k \neq \text{id}$ and $\sigma_k \circ \sigma_k = \text{id}$.
- Fixed field of σ : $\{a \in K : \sigma(a) = a\}$.
- From 9(c): $i = \zeta^2$, $\sqrt{2} = \zeta + \zeta^7$, $K = \mathbb{Q}(i, \sqrt{2})$.

Show that for each $k \in \{3, 5, 7\}$ there is an automorphism σ_k of K with $\sigma_k(\zeta) = \zeta^k$.

YOUR ANSWER question 0001 · attempt 22 of 24 · tap to write

$\sim \forall k \in \{3, 5, 7\}, \exists \sigma_k$ automorph. of $K = \mathbb{Q}(i, \sqrt{2})$

$\therefore \sigma_k(\zeta_8) = \zeta_8^k$

3:
 $\sigma_3(\zeta_8) = \zeta_8^3$

Note ζ_8^3 is its own inverse...

ζ_8^5 and ζ_8^7 are also their
own inverses in $\mathbb{Q}(\zeta_8)$...

I feel
like I
need to
use this
but honestly
don't know what
I'm doing...

LESSON check 9d-existence-recipe

0023

What you noticed is true of the exponents, not of ζ^3 itself, and it is the right instinct. $\zeta^3 \cdot \zeta^3 = \zeta^6 \neq 1$, so ζ^3 is not its own

inverse. What is true is $3 \cdot 3 = 9 \equiv 1 \pmod{8}$, and likewise $5 \cdot 5 \equiv 1$ and $7 \cdot 7 \equiv 1$. So $(\zeta^k)^k = \zeta^{k^2} = \zeta$ for each $k \in \{3, 5, 7\}$.

That fact does two jobs. It is the order-2 piece later: $\sigma_k(\sigma_k(\zeta)) = \zeta^{k^2} = \zeta$. And it is half of existence now.

Existence is the 8(c) recipe, in two facts:

- ζ^k has the same minimal polynomial as ζ , namely $x^4 + 1$. Then 8(a) gives an isomorphism $\mathbb{Q}(\zeta) \rightarrow \mathbb{Q}(\zeta^k)$ with $\zeta \mapsto \zeta^k$.
- $\mathbb{Q}(\zeta^k) = \mathbb{Q}(\zeta)$, so that isomorphism lands in K and is an automorphism. Your observation gives this: $\zeta = (\zeta^k)^k \in \mathbb{Q}(\zeta^k)$, and $\zeta^k \in \mathbb{Q}(\zeta)$.

Fact 2 is done. Fact 1 is the only gap.

Problem 9(d), existence. With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$: for $k \in \{3, 5, 7\}$, show there is an automorphism σ_k of K with $\sigma_k(\zeta) = \zeta^k$.

What you need:

- $\zeta = e^{2\pi i/8}$, so $\zeta^8 = 1$ and $\zeta^4 = -1$.
- $x^4 + 1$ is irreducible over $\mathbb{Q}$ (9(a)), so it is the minimal polynomial of any root of it.
- 8(a): if α, β have the same minimal polynomial over F , there is a unique isomorphism $F(\alpha) \rightarrow F(\beta)$ fixing F with $\alpha \mapsto \beta$.
- An automorphism of K is an isomorphism $K \rightarrow K$.

Show that ζ^k is a root of $x^4 + 1$ for each $k \in \{3, 5, 7\}$.

YOUR ANSWER question 0001 · attempt 23 of 24 · tap to write

~ $\forall k \in \{3, 5, 7\}, \exists \sigma_k$ automorph. of $K = \mathbb{Q}(i, \sqrt{2})$

~ $\sigma_k(\zeta_8) = \zeta_8^k$

$\rightarrow \zeta_8$ is a root of $x^4 + 1$

$\rightarrow$ If the same is true for ζ_8^k , we have

an automorphism ρ of $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ w/

$$\rho(\zeta_8) = \zeta_8^k$$

$$\text{Note } x^4 + 1 \Big|_{x=\zeta_8^3} = (e^{2\pi i \cdot 3/8})^4 + 1 = e^{3\pi i} + 1 = -1 + 1 = 0 \quad (\checkmark)$$

$$\Big|_{x=\zeta_8^5} = e^{5\pi i} + 1 = 0 \quad (\checkmark)$$

$$\Big|_{x=\zeta_8^7} = e^{7\pi i} + 1 = 0 \quad (\checkmark)$$

LESSON check 9d-order-two

0024

Existence is right, for all three k . $(\zeta^k)^4 = e^{k\pi i} = -1$ because k is odd, so ζ^k is a root of $x^4 + 1$. That is fact 1, and fact 2 was already done. So 8(a) gives σ_k , and it lands in K . Existence is now typeset in 9(d) of the worksheet.

One small thing the file fixes for you: what 8(a) hands you is an isomorphism $\mathbb{Q}(\zeta) \rightarrow \mathbb{Q}(\zeta^k)$. It is an automorphism of K only because $\mathbb{Q}(\zeta^k) = K$. The write-up says so in one line.

Next piece of 9(d): order 2.

Problem 9(d), order 2. With $\zeta = \zeta_8$ and $K = \mathbb{Q}(\zeta)$, let σ_k be the automorphism of K with $\sigma_k(\zeta) = \zeta^k$, for $k \in \{3, 5, 7\}$. Show that each σ_k has order 2.

What you need:

- $\zeta = e^{2\pi i/8}$, so $\zeta^8 = 1$, and $\zeta^j = \zeta^{j'}$ exactly when $j \equiv j' \pmod{8}$.
- $k^2 \equiv 1 \pmod{8}$ for $k = 3, 5, 7$ (you noticed this last time).
- An automorphism of $\mathbb{Q}(\zeta)$ fixes $\mathbb{Q}$ and is determined by where it sends ζ .
- σ has order 2 means $\sigma^2 = \text{id}$ and $\sigma \neq \text{id}$.

Show that $\sigma_k^2 = \text{id}$ and $\sigma_k \neq \text{id}$ for each $k \in \{3, 5, 7\}$.

YOUR ANSWER

Show $\forall k \in \{3, 5, 7\}, \exists \sigma_k$ automorph. of $K = \mathbb{Q}(i, \sqrt{2})$

w/ $\sigma_k(\zeta_8) = \zeta_8^k$

$\rightarrow \zeta_8$ is a root of $x^4 + 1$

$\rightarrow$ If the same is true for ζ_8^k , we have an automorphism ρ of $\mathbb{Q}(\zeta_8) = \mathbb{Q}(i, \sqrt{2})$ w/ $\rho(\zeta_8) = \zeta_8^k$

Note $x^4 + 1 \Big|_{x=\zeta_8^3} = (e^{2\pi i \cdot 3/8})^4 + 1 = e^{3\pi i} + 1 = -1 + 1 = 0 \quad (\checkmark)$

$$\Big|_{x=\zeta_8^5} = e^{5\pi i} + 1 = 0 \quad (\checkmark)$$

$$\Big|_{x=\zeta_8^7} = e^{7\pi i} + 1 = 0 \quad (\checkmark)$$